

Design FMEA (DFMEA)

Case ID: VBF-T2R1MIMO-DFMEA-01

Date: 2026-08-30

Type: Qualitative (Virtual Design)

#	Function	Potential Failure Mode	Potential Effect	Severity (1-10)	Potential Cause	Occurrence (1-10)	Current Design Control	Detection (1-10)	RPN	Recommended Action
1	Energy storage	Insufficient energy density	Below 327.18 Wh/kg target	7	Electrode too thin / too porous	2	ED verified at 378.69 Wh/kg (margin: 15.7%)	1	14	Monitor electrode thickness tolerance
2	Fast charge	Lithium plating at 4C	Cell degradation, safety hazard	9	Insufficient electrolyte transport	2	Enhanced electrolyte (D=2.0e-9, σ=2.5), verified min V=+0.0006V	1	18	Validate electrolyte formulation experimentally
3	Fast charge	Excessive temperature rise	Thermal degradation	7	High current density, poor thermal management	2	Thermal lumped model verified T_max=344.95K (<350K)	2	28	Add thermal management system in pack design
4	Low-temp operation	Capacity loss at -20°C	Reduced runtime	5	Electrolyte transport limitation	1	Verified 99.99% retention (enhanced transport)	1	5	None (exceeds target)
5	Long cycling	Excessive SEI growth	Capacity fade, impedance rise	6	High SEI formation rate	3	SEI rate reduced 71× (7e-15 m/s), verified 518nm@500cyc	2	36	Validate SEI suppression mechanism experimentally
6	Calendar aging	SEI growth during storage	Capacity fade	5	Continuous SEI formation	3	Same SEI suppression applies	3	45	Long-term storage test recommended

RPN Summary: Highest RPN = 45 (calendar aging SEI). All RPNs < 100 (acceptable for virtual design stage).
Key Risk: SEI suppression relies on estimated electrolyte/additive performance — requires experimental validation.